

$$\underline{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \underline{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} \in \mathbb{R}^n \quad \underline{q}^T \underline{X} = \begin{bmatrix} q_1 & q_2 & \dots & q_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = q_1 x_1 + q_2 x_2 + \dots + q_n x_n$$

$$(\underline{q}^T \underline{X})^2 = (q_1 x_1 + q_2 x_2 + \dots + q_n x_n)^2 = q_1^2 x_1^2 + q_2^2 x_2^2 + \dots + q_n^2 x_n^2 + 2q_1 q_2 x_1 x_2 + \dots + 2q_1 q_n x_1 x_n + \dots + 2q_{n-1} q_n x_{n-1} x_n$$

$$(\underline{q}^T \underline{X})(\underline{q}^T \underline{X}) = (\underline{q}^T \underline{X})(\underline{X}^T \underline{q}) = \underline{q}^T (\underline{X} \underline{X}^T) \underline{q}$$

$$\underline{X}^T \underline{q} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} = q_1 x_1 + q_2 x_2 + \dots + q_n x_n$$

$$\underline{X} \underline{X}^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} = \begin{bmatrix} x_1^2 & x_1 x_2 & \dots & x_1 x_n \\ x_1 x_2 & x_2^2 & \dots & x_2 x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_1 x_n & x_2 x_n & \dots & x_n^2 \end{bmatrix} = [x_i x_j]_{\substack{i=1, \dots, n \\ j=1, \dots, n}}$$

$$R_{\underline{X}} = E[\underline{X} \underline{X}^T] = [E[x_i x_j]]_{\substack{i=1, \dots, n \\ j=1, \dots, n}} = [cov(x_i, x_j)]_{\substack{i=1, \dots, n \\ j=1, \dots, n}}$$

$\underline{\mu}_X = E[\underline{X}] = \begin{bmatrix} E[x_1] \\ E[x_2] \\ \vdots \\ E[x_n] \end{bmatrix}$

valore delle medie

$$(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T = \left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} \right) \begin{bmatrix} \dots \end{bmatrix}^T$$

$$= \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \\ \vdots \\ x_n - \mu_n \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 & x_2 - \mu_2 & \dots & x_n - \mu_n \end{bmatrix} = \begin{bmatrix} (x_1 - \mu_1)^2 & (x_1 - \mu_1)(x_2 - \mu_2) & \dots & (x_1 - \mu_1)(x_n - \mu_n) \\ (x_1 - \mu_1)(x_2 - \mu_2) & (x_2 - \mu_2)^2 & \dots & (x_2 - \mu_2)(x_n - \mu_n) \\ \vdots & \vdots & \ddots & \vdots \\ (x_1 - \mu_1)(x_n - \mu_n) & (x_2 - \mu_2)(x_n - \mu_n) & \dots & (x_n - \mu_n)^2 \end{bmatrix}$$

$$\underline{C}_{\underline{X}} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T] = [cov(x_i, x_j)]_{\substack{i=1, \dots, n \\ j=1, \dots, n}} = [(\underline{x}_i - \underline{\mu}_i)(\underline{x}_j - \underline{\mu}_j)]_{\substack{i=1, \dots, n \\ j=1, \dots, n}}$$

$$= \begin{bmatrix} \sigma_1^2 & cov(x_1, x_2) & \dots & cov(x_1, x_n) \\ cov(x_2, x_1) & \sigma_2^2 & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ cov(x_n, x_1) & cov(x_n, x_2) & \dots & \sigma_n^2 \end{bmatrix}$$

$$= \begin{pmatrix} \text{cov}(x_1, x_1) & \dots & \text{cov}(x_1, x_n) \\ \vdots & \ddots & \vdots \\ \text{cov}(x_n, x_1) & \dots & \text{cov}(x_n, x_n) \end{pmatrix}$$

$$0 \leq \mathbb{E}[(q^T \underline{x})^2] = \mathbb{E}\{q^T \underline{x} \underline{x}^T q\} = q^T \mathbb{E}\{\underline{x} \underline{x}^T\} q = q^T R_x q \geq 0 \quad \forall q$$

$\underline{x}$

$$\underline{y} = \underline{x} - \underline{\mu}$$

$$0 \leq \mathbb{E}[(q^T \underline{y})^2] = \mathbb{E}\{q^T (\underline{x} - \underline{\mu}) (\underline{x} - \underline{\mu})^T q\} = q^T \mathbb{E}\{(\underline{x} - \underline{\mu}) (\underline{x} - \underline{\mu})^T\} q = q^T C_x q \quad \forall q$$

$\forall q$

$$C_x = \mathbb{E}\{(\underline{x} - \underline{\mu}) (\underline{x} - \underline{\mu})^T\} = \mathbb{E}\{\underline{x} \underline{x}^T - \underline{\mu} \underline{x}^T - \underline{x} \underline{\mu}^T + \underline{\mu} \underline{\mu}^T\}$$

$$= \mathbb{E}\{\underline{x} \underline{x}^T\} - \underline{\mu} \mathbb{E}\{\underline{x}^T\} - \mathbb{E}\{\underline{x}\} \underline{\mu}^T + \underline{\mu} \underline{\mu}^T$$

$$= R_x - \underline{\mu} \underline{\mu}^T$$

$X_0(n, \omega)$

$X_0(n_1) \quad X_0(n_2)$

$\dots$
i.e. $\forall n_1, n_2$

$$\underbrace{X_0(n_1) \quad X_0(n_2) \quad \dots \quad X_0(n_M)}_{\text{i.e. } \forall (n_2, n_2, \dots, n_M)} \quad \forall M$$

$$f_{X_0}(d_1, d_2, \dots, d_M; n_1, n_2, \dots, n_M) = f_{X_0}(d_1; n_1) f_{X_0}(d_2; n_2) \cdot \dots \cdot f_{X_0}(d_M; n_M)$$

$$f_{X_0}(d)$$

$$= \prod_{i=1}^M f_{X_0}(d_i; n_i) \quad \text{i.e. } \text{IP} \in \text{ensemble}$$

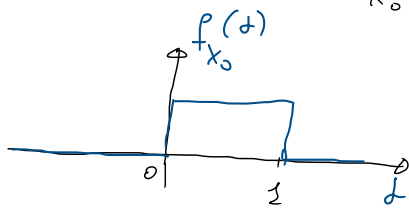
$$f_{X_0}(d; n)$$

è indep da $n \Leftrightarrow$ SSS di ordine 1

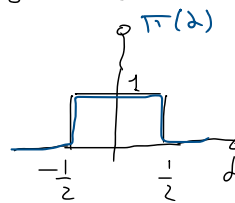
IDENTICAMENTE
DISTRIBUZIONE

$$X_0 \text{ è IID } \Leftrightarrow f_{X_0}(d_1, d_2, \dots, d_M; n_1, n_2, \dots, n_M) = f_{X_0}(d_1) f_{X_0}(d_2) \cdot \dots \cdot f_{X_0}(d_M) = \prod_{i=1}^M f_{X_0}(d_i)$$

$$X_0 \text{ e IID } \Rightarrow f_{X_0}(d_1, d_2, \dots, d_n; n_1, n_2, \dots, n_n) = f_{X_0}(d_1) f_{X_0}(d_2) \dots f_{X_0}(d_n) = \prod_{i=1}^n f_{X_0}(d_i)$$



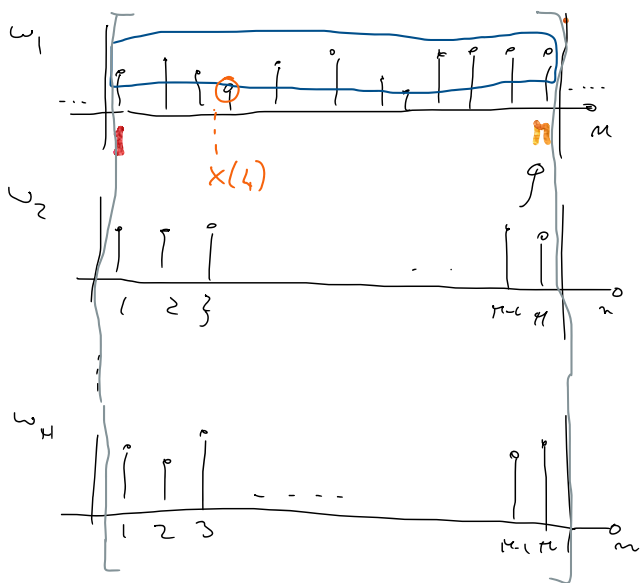
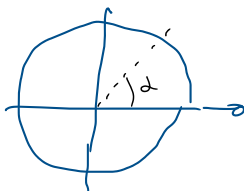
$$f_{X_0}(d) = \pi(d - \frac{1}{2})$$



$$\frac{d}{2\pi}$$

$$\frac{d}{3\pi^0}$$

$$d \in (0, 2\pi)$$



$$X = \text{rand}(1, M);$$

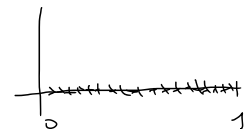
$$Xc = \text{rand}(N, M);$$

$$X2 = Xc(2, :);$$

$$X25 = X(2, 3:7); = X2(3:7);$$

pseudo-random

$$f_{X_0}(d) = \pi(d - \frac{1}{2})$$



$$P[X_0(n) \in (a, b)] = \int_a^b f_{X_0}(d) dd = \int_a^b dd = b - a$$

$$0 < a < b < 1$$

IID random dnm

$$\begin{cases} X = \text{rand}(N, 1); \\ Xc = \text{rand}(N, 10); \\ X = Xc(:, 10); \end{cases}$$

$$X = \text{rand}(1, N);$$

$$P_r[X_0(n) \leq b] = b$$

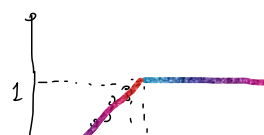
$$0 < b < 1$$

0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0

$$(X \leq b)$$

$$\frac{\text{sum}(X \leq b)}{N}$$

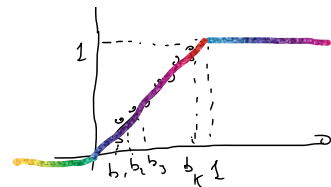
$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$



$$0 < X_1 < 0.1$$

$$0 < X_1 < 0.1$$

N



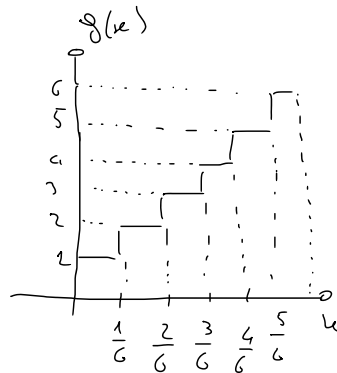
$$I = \text{ifind}(x \leq 0.1)$$

escher $\xrightarrow{\text{euler comp}}$

$$y = x(I+1)$$

$$x = \text{rand}(N, M);$$

$$Y(n) = g(X_o(n))$$



$$\begin{cases} 1 & \text{con prob. } 1/6 \\ 2 & \text{con prob. } 1/6 \\ \vdots & \\ 6 & \text{con prob. } 1/6 \end{cases}$$

$$Y=1 \Leftrightarrow 0 < X_o < \frac{1}{6}$$

$$P[Y=1] = P[X_o < \frac{1}{6}] = \frac{1}{6} \quad \text{ceil}(6 * X_o)$$

$$Y=2 \Leftrightarrow \frac{1}{6} < X_o < \frac{2}{6}$$

$$P[Y=2] = P[\frac{1}{6} < X_o < \frac{2}{6}] = \frac{1}{6}$$

$$Y=3 \Leftrightarrow \frac{2}{6} < X_o < \frac{3}{6}$$

$$P[Y=3] = P[\frac{2}{6} < X_o < \frac{3}{6}] = \frac{1}{6}$$

$$Y=4 \Leftrightarrow \frac{3}{6} < X_o < \frac{4}{6}$$

$$P[Y=4] = P[\frac{3}{6} < X_o < \frac{4}{6}] = \frac{1}{6}$$

$$Y=5 \Leftrightarrow \frac{4}{6} < X_o < \frac{5}{6}$$

$$P[Y=5] = P[\frac{4}{6} < X_o < \frac{5}{6}] = \frac{1}{6}$$

$$Y=6 \Leftrightarrow \frac{5}{6} < X_o < 1$$

$$P[Y=6] = P[\frac{5}{6} < X_o < 1] = \frac{1}{6}$$

$$x_c = \text{rand}(N, M);$$

$$M=2$$

$$D_c = \text{ceil}(6 * x_c);$$

$$D_s = D_c(:, 1) + D_c(:, 2)$$

$$\begin{bmatrix} 2 & 4 \\ 5 & 1 \\ 4 & 3 \\ \vdots & \vdots \end{bmatrix}$$

$$\begin{bmatrix} 8 \\ 6 \\ 7 \\ \vdots \end{bmatrix}$$

$$\begin{bmatrix} 7 \\ 5 \\ 4 \\ \vdots \end{bmatrix}$$

$$\begin{bmatrix} 6 \\ 1 \\ 3 \\ \vdots \end{bmatrix}$$

for $i=3, \dots, 12$
2, ...

$$\text{sum}(D_s = 7)$$

N

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$



$$\alpha = \frac{1}{2}$$

$$P[D=6] = 4$$

$$P[D=i] = \frac{1-i}{5}$$

$$x = \text{rand}(N, 1)$$

$$y = \text{ceil}(10 * x);$$

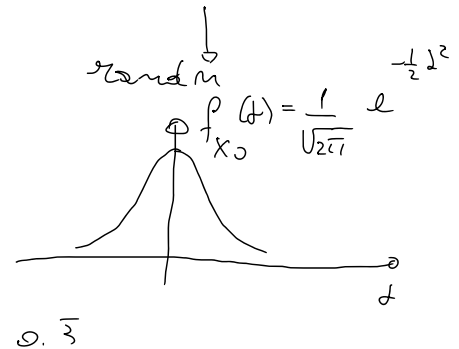
$$y(y > 6) = 6;$$

$$\begin{cases} im = \text{find}(y > 6); \\ y(im) = 6; \end{cases}$$

$$D_{od} = [D_e(N, 1), y];$$

$$D_s = D_{od}(:, 1) + D_{od}(:, 2);$$

$$\frac{1}{3}$$



Luglio 2025

10 — 5 "1"
3 "2"
2 "3"

$x_1 \backslash x_2$	1	2	3	
1	$\frac{1}{2} \cdot \frac{4}{9}$	$\frac{1}{2} \cdot \frac{3}{9}$	$\frac{1}{2} \cdot \frac{2}{9}$	$\frac{1}{2} = \frac{5}{10}$
2	$\frac{3}{10} \cdot \frac{5}{9}$	$\frac{3}{10} \cdot \frac{2}{9}$	$\frac{3}{10} \cdot \frac{2}{9}$	$\frac{3}{10}$
3	$\frac{2}{10} \cdot \frac{5}{9}$	$\frac{2}{10} \cdot \frac{3}{9}$	$\frac{2}{10} \cdot \frac{1}{9}$	$\frac{2}{10}$

$$P[X_1=1] = \frac{5}{10} = \frac{1}{2}$$

$$P[X_1=2] = \frac{3}{10}$$

$$P[X_1=3] = \frac{2}{10}$$

$$P[X_1=1, X_2=1] = P[X_1=1] P[X_2=1 | X_1=1] = \frac{1}{2} \cdot \frac{4}{9}$$

$$P[X_1=1] = \sum_{i=1}^3 P[X_1=1, X_2=i]$$

$$P[X_2=3] = \sum_{i=1}^3 P[X_1=i, X_2=3]$$

$$P[X_2=1] = \frac{2}{9} + \frac{15}{90} + \frac{10}{90} = \frac{20+15+10}{90} = \frac{45}{90} = \frac{1}{2}$$

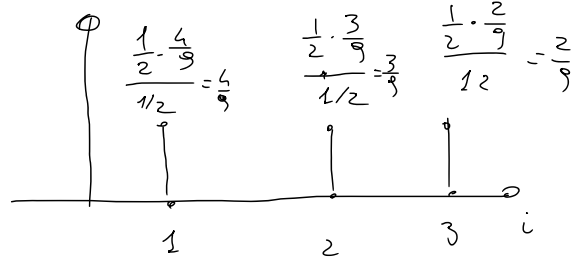
$$P[X_2=2] = \frac{3}{18} + \frac{6}{90} + \frac{6}{90} = \frac{15+6+6}{90} = \frac{27}{90} = \frac{3}{10}$$

$$E[X_1] = E[X_2] = 1 \times \frac{1}{2} + 2 \times \frac{3}{10} + 3 \times \frac{2}{10} = \frac{5+6+6}{10} = \frac{17}{10} = 1.7$$

$$E[X_2 | X_1=1]$$

$$P[X_2=i | X_1=1] = \frac{P[X_2=i, X_1=1]}{P[X_1=1]} = \frac{P_{X_2|X_1=1}(i)}{P[X_1=1]}$$

$$P[X_2 = i | X_1 = 1] = \frac{P[X_1 = 1, X_2 = i]}{P[X_1 = 1]} = P[X_2 = i | X_1 = 1]$$



$$E[X_2 | X_1 = 1] = 1 \times \frac{4}{9} + 2 \times \frac{3}{9} + 3 \times \frac{2}{9} = \frac{4+6+6}{9} = \frac{16}{9}$$

$$M = \begin{bmatrix} x_1 & x_2 \\ 1 & 0 \\ 2 & 0 \\ 3 & 0 \\ 1 & 0 \\ 2 & 0 \\ 3 & 0 \\ 1 & 0 \\ 2 & 0 \\ 3 & 0 \end{bmatrix}$$

$$\text{mean} \left[\begin{matrix} 1 \\ 1 \end{matrix} \right]$$

$$im = \text{find}(M(:, 1) == 1);$$

$$J_2 = M(im, 2);$$

$$\text{mean}(J_2)$$

10 "1"
6 "2"
4 "3"

$X_1 \backslash X_2$	1	2	3
1	$\frac{1}{2} \cdot \frac{9}{19}$	$\frac{1}{2} \cdot \frac{6}{19}$	$\frac{1}{2} \cdot \frac{4}{19}$
2	$\frac{6}{20} \cdot \frac{10}{19}$	$\frac{6}{20} \cdot \frac{5}{19}$	$\frac{6}{20} \cdot \frac{4}{19}$
3	$\frac{4}{20} \cdot \frac{10}{19}$	$\frac{4}{20} \cdot \frac{6}{19}$	$\frac{4}{20} \cdot \frac{3}{19}$